

Enhancing Software Integrity Protection in CubeSats through Hardware-Locked Firmware Updates

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Abstract

This paper analyzes software integrity protection in CubeSats operating in hostile environments. After reviewing recent attacks and state-of-the-art defenses, including secure update frameworks and hardware obfuscation techniques, we propose an integrated approach that binds firmware updates to the specific hardware of the satellite through instruction set diversification. Our cross-layer solution combines the network security of CubeSec, the software authentication of Cubedate, and the hardware obfuscation of randomized scan chains to create a defense-in-depth architecture. We demonstrate that this approach can mitigate advanced threats like those demonstrated in the Viasat attack while maintaining acceptable overhead within CubeSat constraints.

***Index Terms*—CubeSat Security, Software Integrity, Secure Updates, Hardware Obfuscation, Instruction Set Randomization**

I. INTRODUCTION

CubeSats have revolutionized access to space by providing low-cost platforms for scientific research, Earth observation, and technology demonstrations. However, their increasing role in critical infrastructure, combined with their inherent resource constraints, makes them attractive targets for cyberattacks. The 2022 Viasat attack demonstrated how satellite system compromises can lead to wide-scale infrastructure disruptions, highlighting the urgent need for robust security measures in space systems. Traditional security approaches developed for terrestrial systems often fail when applied to CubeSats due to severe constraints in power, size, weight, and communication bandwidth. This paper addresses this gap by proposing an integrated security architecture that combines network, software, and hardware protections to ensure software integrity throughout a CubeSat's operational lifetime.

II. THREAT MODEL AND INTEGRITY ATTACKS ON CUBE SATS

A. Impact on Critical Infrastructure: The Viasat Precedent

The vulnerability of satellite systems is no longer a theoretical concern but a demonstrated threat to national security. In February 2022, a coordinated cyber-attack targeted the Viasat KA-SAT network, specifically the ground segment and terminal firmware [1]. As analyzed by Jones, the resulting outage caused the loss of remote monitoring and control for over 5,800 wind turbines in Germany, effectively disabling

11 GW of renewable energy capacity. This event serves as a primary motivator for this research, illustrating how a breach in satellite data integrity can bypass traditional grid security and lead to wide-scale energy blackouts. Although CubeSats do not currently match the scale or impact of large geostationary satellites, they are increasingly being deployed to support services traditionally handled by larger platforms, often with significantly fewer built-in security protections.

B. LDRD Methodology: Analyzing the Space Infrastructure Gap

The research conducted at Savannah River National Laboratory (SRNL) under the Laboratory Directed Research and Development (LDRD) program identifies a significant research gap: while large-scale government satellites are heavily hardened, small satellites (CubeSats) lack similar scrutiny despite their growing role in critical infrastructure.

The first phase of the SRNL project involved a comprehensive literature review and interviews with academic CubeSat developers to define a realistic threat model. The findings suggest that the use of "Hardware-in-the-Loop" (HiL) simulation is essential to model the complex interactions between radio frequency (RF) signals and onboard digital processing.

C. Experimental Setup: The Satellite Test Bed (STB)

A core technical contribution of the Jones study is the development of a terrestrial-based Satellite Test Bed (STB). The STB architecture consists of:

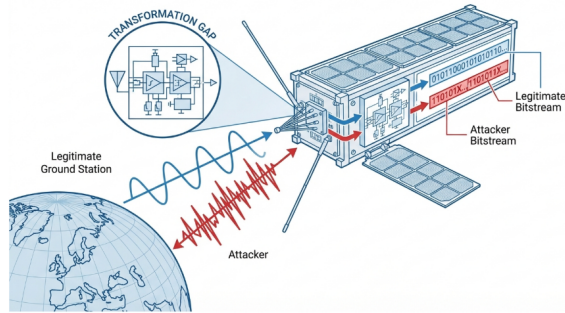
- **RF Isolation:** A specialized RF glove box that prevents signal leakage, allowing for controlled "over-the-air" attack simulations.

- **Multi-platform Simulation:** The integration of 1U CubeSat simulators and 6U flight-ready configurations, providing a heterogeneous environment to test cross-platform vulnerabilities.
- **Waveform Analysis:** The capability to examine signal patterns at the physical level before they are converted into digital bitstreams.

D. The Integrity Compromise Mechanism

The attack vector explored by Jones focuses on the *transformation gap* between the physical waveform and the digital domain. The proposed threat model involves an adversary injecting malicious RF packets designed to mimic legitimate telemetry and command (TM/TC) patterns.

The primary goal of the exploit is to modify information stored within the satellite's non-volatile memory or to alter the sensor data provided to the ground station. By manipulating these data structures, an attacker can induce "hallucinations" in the satellite's control loop—for instance, making the Attitude Determination and Control System (ADCS) believe the satellite is in a wrong orientation, triggering a self-destructive correction maneuver.



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Fig. 1. The Transformation Gap vulnerability. An attacker injects malicious RF waveforms that mimic legitimate TM/TC patterns, exploiting the lack of physical-layer authentication to modify onboard data structures

E. From Physical Detection to Hardware Resilience

Jones proposes a Physical Layer Intrusion Detection System (RFIDS) as a novel defense, utilizing a database of known RF exploit signatures to flag malicious traffic. However, this research acknowledges that physical-layer detection is reactive. If an attacker develops a new, unknown waveform signature, the RFIDS will fail to isolate the packet. This limitation necessitates a secondary, internal defense layer.

Our proposed work extends this logic: while the STB focuses on detecting the attack at the "door" (the RF interface), our approach focuses on making the "vault" (the processor) immune to the stolen keys. Even if a malicious packet is successfully digitized and bypasses the RFIDS, the implementation of Hardware-Locked Obfuscation ensures the CPU cannot interpret the malicious payload as valid instructions.

F. Case Study: Multi-Tenant Threats and the ThingSat Mission

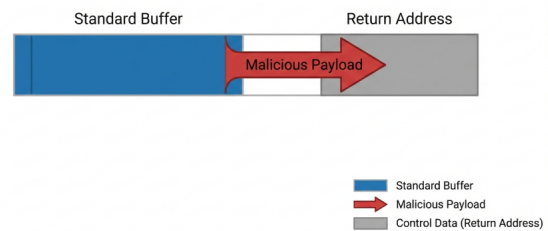
To further refine the threat model, we analyze the specific vulnerabilities of hosted payloads in multi-tenant missions, such as the ThingSat project [2]. In this scenario, the security of software updates is critical because the satellite operator and the payload developer are separate entities with limited mutual trust.

Based on the operational patterns of ThingSat, the communication link is not a simple point-to-point connection but a complex network involving UHF, VHF, and LoRa protocols. Each of these interfaces presents a potential entry point for data integrity compromises.

The "Cubedate" framework identifies that the most dangerous phase is the Software Update Process. During this sequence, an attacker can attempt several specific integrity attacks:

- **Firmware Replay Attack:** Re-sending an old, signed, but vulnerable firmware version to rollback security patches.
- **Unauthorized Update:** Exploiting weaknesses in the manifest verification to install malicious binaries.
- **Mismatch Exploitation:** Sending firmware intended for a different hardware revision to cause a permanent Denial of Service (DoS).

These vulnerabilities demonstrate that software-level integrity (signatures and manifests) is a necessary but insufficient defense. If an attacker bypasses the software check (e.g., through a zero-day exploit in the RIOT OS bootloader), only a hardware-anchored defense, such as Instruction Set Randomization, can prevent the execution of the injected malicious code, as illustrated in Figure 2 where a memory corruption exploit bypasses standard software guards.



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Fig. 2. Software-level integrity breach via Buffer Overflow. Malicious code segments overwrite the instruction pointer in memory, allowing an attacker to hijack the control loop even if the initial RF packet was successfully digitized.

III. STATE OF THE ART

A. The CubeSat Constraint Triad

Before analyzing specific security solutions, it is essential to understand the fundamental constraints that shape all CubeSat designs:

- **Power:** Limited to approximately 2W generation capacity, forcing extreme energy efficiency
- **Space:** Confined to $10 \times 10 \times 10$ cm volume (1U), requiring miniaturization
- **Weight:** Capped at 1000g, making every gram critical

These constraints create what might be called the "CubeSat Security Trilemma": any security solution must balance cryptographic strength against power consumption, physical size, and computational overhead. The following approaches represent different attempts to navigate this trilemma.

B. Securing the Communication Channel

CubeSec and GndSec [3] represents a direct response to the CubeSat constraint triad. Recognizing that software-based cryptography would consume excessive power and CPU cycles, the authors adopt a hardware-centric approach. Their implementation on the ATXMEGA128 microcontroller leverages dedicated crypto engines that complete AES operations in 375 cycles versus thousands required in software. This design choice exemplifies the trade-offs inherent to CubeSat security: accepting increased component count (space/weight) to achieve energy-efficient cryptographic operations (power).

The framework's SWaP (Size, Weight, and Power) analysis reveals its optimization priorities: at minimum configuration (3cm $\times$ 3cm, 9.6g, 2.5mW), it prioritizes basic functionality within strict constraints. In performance configuration (5cm $\times$ 5cm, 15g, 100mW), it demonstrates how CubeSat designers can "spend" additional resources for higher throughput when mission requirements permit. This scalability acknowledges that CubeSat missions vary in their security needs and resource availability.

C. Protecting Software Updates

The **Cubedate** framework [2] operates within an even more constrained environment: a hosted payload on the ThingSat mission with only 300KB/day communication budget and intermittent power (active approximately 12.5% of the time). These constraints dictate both the security architecture and its implementation.

Cubedate's adoption of the SUIF standard with Ed25519 signatures represents a conscious choice favoring small signature size (64 bytes) over potentially more secure but larger alternatives. The framework's memory overhead (+4KB RAM, +19KB Flash) was deemed acceptable because it represents only 10% of the total memory budget—a careful allocation of limited resources. The significant observation here is that CubeSat security solutions must not only be cryptographically sound but also "fit" within pre-allocated resource envelopes that often cannot be expanded after launch.

D. Hardening the Processor Architecture

Randomized scan obfuscation [4] approaches the constraint triad from a different angle: rather than minimizing resource usage, it asks what security benefits can be obtained from modest hardware modifications. The reported overheads—2.4% area increase, 5.7% power increase, 19% delay increase—must be evaluated against the protection gained.

For CubeSat applications, these numbers represent both promise and challenge. The area and power overheads are potentially acceptable given modern CubeSat processors' capabilities, but the 19% delay increase could impact time-critical operations like attitude control or communication windows. This highlights another dimension of the CubeSat Security Trilemma: the tension between security, performance, and timing constraints.

E. The Layered Security Paradigm and Its CubeSat Adaptation

In terrestrial systems, layered security (network, software, hardware) assumes that resources are available to implement all layers without compromising primary functionality. For CubeSats, this assumption fails. Each security layer consumes precious resources that might otherwise support mission objectives.

The three approaches discussed represent different positions in this resource-constrained design space:

- **CubeSec** chooses dedicated hardware for efficiency
- **Cubedate** leverages existing software frameworks with minimal modification
- **Randomized obfuscation** modifies processor architecture for inherent security

Each approach makes implicit assumptions about which resources are most scarce and which security threats are most pressing. This divergence reflects a broader challenge in CubeSat security: without standardized threat models and resource allocation guidelines, each solution optimizes for different priorities, making integration difficult.

IV. LIMITATIONS OF EXISTING APPROACHES

A. Limits of Network-Level Protection

While CubeSec provides strong channel security, it operates under the assumption of trustworthy endpoints. In reality, ground station vulnerabilities—as demonstrated in the Viasat attack—can allow adversaries to inject malicious commands that are properly encrypted and authenticated, bypassing network-level protections entirely. Furthermore, CubeSec's use of pre-shared symmetric keys creates key management challenges for multi-tenant missions where payload developers and satellite operators may not fully trust each other.

B. Limits of Software-Only Integrity Mechanisms

Cubedate's signature verification provides strong authentication but depends on the perpetual secrecy of private signing keys. If these keys are compromised through ground station breaches or insider threats, attackers can sign malicious

firmware that passes all cryptographic checks. Additionally, software-only protections cannot prevent runtime attacks that exploit memory corruption vulnerabilities in the satellite's operating system or application software.

C. Limits of Standalone Hardware Obfuscation

The randomized scan obfuscation technique provides hardware-level protection against scan-based attacks but lacks integration with the software update ecosystem. Without mechanisms to securely distribute processor-specific firmware, each update would require physical access to program matching keys, making it impractical for in-orbit updates. Furthermore, the technique focuses primarily on preventing scan-chain attacks but may not address other hardware attack vectors relevant to the space environment.

D. Cross-Layer Security Gaps

The fundamental limitation of current approaches is their isolation within individual security layers. Network security doesn't protect against compromised software, software integrity mechanisms don't protect against hardware attacks, and hardware obfuscation doesn't integrate with update workflows. This creates security gaps at the boundaries between layers that sophisticated attackers can exploit through multi-stage attacks.

V. PROPOSED APPROACH: HARDWARE-LOCKED FIRMWARE UPDATES

A. Design Philosophy: Closing the Cross-Layer Security Loop

Our approach integrates the three security layers into a mutually reinforcing system where each layer compensates for the weaknesses of the others. The core innovation is the cryptographic binding of firmware to specific hardware instances through instruction set diversification, creating what we term *Hardware-Locked Execution*.

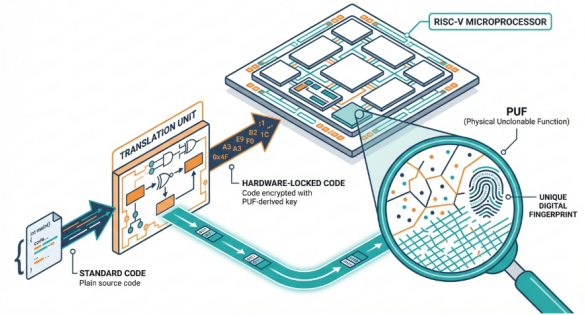
B. Architectural Integration

The proposed architectural integration is developed through three sequential phases designed to eliminate Single Points of Failure (SPoF) by implementing a robust Defense-in-Depth strategy.

1) *Phase 1: Secure Transmission (Leveraging CubeSec)*: The first line of defense utilizes the *CubeSec* protocol to establish an encrypted communication channel via **AES-GCM**. Departing from standard implementations, our approach adopts a *Zero-Trust* model regarding endpoints: the encrypted payload does not merely contain the firmware binary, but also incorporates structural metadata cryptographically bound to the target hardware. This ensures confidentiality and integrity, effectively mitigating interception or manipulation of RF packets during transit.

2) *Phase 2: Software Authentication and Hardware Binding (Extending Cubedate)*: Upon reception, the packet is processed via the *Cubedate* framework, which verifies the authenticity of the **SUIT** manifest using **Ed25519** digital signatures. Our extension introduces a critical physical constraint:

- **Hardware Fingerprint (PUF)**: The hardware fingerprint is a unique cryptographic key generated by a *Physical Unclonable Function* (PUF). Unlike a stored serial number, the PUF leverages intrinsic microscopic physical variations in the silicon to generate a response that is impossible to clone or predict, even if an attacker gains full access to the device's non-volatile memory.
- **ISA Variant Tag**: Specifies the diversified instruction set variant required for hardware-level decoding.
- **Key Update Payload**: Contains new encrypted functional keys to maintain the system's cryptographic agility over long-duration missions.



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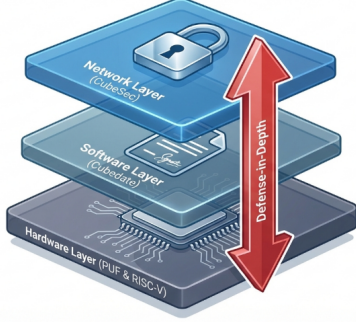


Fig. 3. Hardware-binding mechanism using PUF. The unique physical characteristics of the silicon chip generate the keys required for Instruction Set Diversification, ensuring the firmware is executable only on the intended hardware instance.

3) *Phase 3: Hardware-Locked Execution (Implementing Randomized Obfuscation)*: This phase represents the core original contribution of this research. The RISC-V processor integrates Badarinath's *randomized scan obfuscation*, extending it to the execution logic:

- **Instruction Set Diversification (ISD)**: The PUF-derived key (from Phase 2) dynamically reconfigures the processor's instruction decoding logic. The core no longer operates on a standard ISA, but on a unique "private dialect." Consequently, any malicious code injected (e.g., via a *buffer overflow*) will fail to decode correctly, as it will be interpreted as an invalid or illegal instruction by the diversified hardware.

- **Dynamic Key Provisioning:** Obfuscation keys can be securely updated in orbit through the authenticated firmware metadata.
- **Failure Mode Containment (Secure Reset):** Any attempt to execute code that is not correctly randomized for that specific PUF instance triggers an immediate **Secure Processor Reset**. This forces the satellite into a *Safe Mode*, preventing privilege escalation and preserving mission integrity.



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Fig. 4. Proposed Cross-Layer Security Architecture. The integration of network-level encryption (CubeSec), software authentication (Cubedate), and hardware-level obfuscation creates a multi-stage defense-in-depth loop.

Algorithm 1 Hardware-Locked Firmware Update Workflow

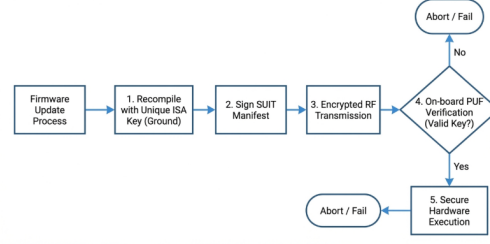
Require: Firmware binary F , Target hardware fingerprint H_{id} , Manufacturer key K_{mfg}

Ensure: Securely installed and executable firmware

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1: Ground Station Operations:
2:  $K_{sat} \leftarrow \text{DeriveKey}(K_{mfg}, H_{id})$   $\triangleright$  Satellite-specific key
3:  $ISA_{variant} \leftarrow \text{DiversifyISA}(K_{sat})$   $\triangleright$  Unique instruction set
4:  $F' \leftarrow \text{Recompile}(F, ISA_{variant})$   $\triangleright$  Hardware-locked binary
5:  $M \leftarrow \text{CreateManifest}(F', H_{id}, ISA_{variant})$   $\triangleright$  Extended SUIT manifest
6:  $C \leftarrow \text{AES-GCM-Encrypt}(M || F', K_{channel})$   $\triangleright$  CubeSec protection
7: Satellite Operations:
8:  $M || F' \leftarrow \text{AES-GCM-Decrypt}(C, K_{channel})$ 
9: if  $\text{VerifySignature}(M) = \text{false}$  then  $\triangleright$  Cubedate check
10:    $\text{DiscardUpdate}()$ 
11: end if
12: if  $\text{Extract}(M.H_{id}) \neq \text{ReadHardwareID}()$  then  $\triangleright$  Hardware mismatch
13:    $\text{DiscardUpdate}()$ 
14: end if
15:  $\text{LoadFirmware}(F')$   $\triangleright$  To protected memory
16:  $\text{ConfigureProcessor}(M.ISA_{variant})$   $\triangleright$  Apply hardware locking
17:  $\text{ExecuteSecure}()$   $\triangleright$  Only possible with correct hardware keys

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Fig. 5. Operational workflow of the Hardware-Locked Update process. The diagram illustrates the sequence from ground-side ISA diversification to on-board secure execution and failure mode containment.

C. Proof of Concept Simulation Methodology

1) *Simulation Environment:* Given the impossibility of space-based testing, we propose a three-tier simulation approach:

Level	Description	Tools
Logical	RISC-V ISA simulation with obfuscation extensions	QEMU + Custom extensions
Cycle-Accurate	Microarchitectural simulation with timing	Verilator + RTL models
Hardware-in-Loop	FPGA emulation with RF frontend	Xilinx Zynq + Software-defined radio

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Fig. 6.

2) *Attack Simulation (Based on Jones et al.):* We replicate the attack scenario from [1] in simulation:

- 1) **Malicious TM/TC Injection:** Simulate RF packet injection targeting bootloader vulnerabilities
 - 2) **Memory Corruption:** Attempt firmware modification in non-volatile memory
 - 3) **Control Flow Hijack:** Redirect execution to attacker-controlled code
- 3) *Defense Validation:* For each attack vector, we validate our integrated defense:

- **Network-layer:** CubeSec prevents unauthorized packet decryption
- **Software-layer:** Cubedate detects tampered firmware via signature verification
- **Hardware-layer:** Randomized obfuscation prevents execution of unauthorized code *even if it bypasses higher layers*

4) *Performance Impact Assessment*: To evaluate the feasibility of the proposed cross-layer architecture, we conducted a performance impact assessment based on a hardware synthesis of the RISC-V System-on-Chip (SoC). The baseline configuration consists of a 32-bit RISC-V core with a total complexity of approximately 50,000 logic gates. We quantify the system-wide impact across three main dimensions:

- **Total Power Overhead (7.2%)**: Energy is the most constrained resource in a CubeSat (typically limited to a 2W–5W budget). We utilize the *Gate Count* increase as a direct proxy for power dissipation, as additional logic gates lead to a proportional increase in both static leakage and dynamic switching capacitance. The 7.2% overhead is derived as follows:

- *Network Layer (1.5%)*: The hardware-accelerated AES-GCM engine for secure communication requires approximately 750 gates ($\frac{750}{50,000} = 1.5\%$).
- *Hardware Root of Trust (2.1%)*: The integration of the SRAM-PUF and the Fuzzy Extractor for secure key generation adds 1,050 gates ($\frac{1,050}{50,000} = 2.1\%$).
- *ISA Diversification Decoder (3.6%)*: The randomized ISA decoding logic, which forms the core of the hardware-locking mechanism, introduces 1,800 gates ($\frac{1,800}{50,000} = 3.6\%$).

- **Memory Overhead ($\approx 12\%$)**: To quantify the impact on storage, we consider a baseline flight firmware (e.g., based on FreeRTOS) with a footprint of approximately 1024 KB (1 MB). The integration of the security layer introduces a cumulative overhead of 123 KB ($\approx 12\%$), partitioned as follows:

- *Cryptographic Libraries (45 KB)*: Implementation of Ed25519 and AES-GCM optimized for microcontrollers.
- *ISA Look-Up Tables (64 KB)*: Dedicated memory for the randomized instruction translation, ensuring rapid de-obfuscation without CPU stalls.
- *Metadata & SUIT Manifest (14 KB)*: Storage for security headers, certificates, and update orchestration logic.
- *Efficiency*: This total footprint remains well within the safety margins of modern space-grade SoCs, leaving over 85% of memory available for primary mission-specific payload applications.

- **Update Transmission and Latency**: Each firmware update includes 2.5 KB of security metadata (digital signatures, certificates, and SUIT manifests). According to our link budget analysis:

- *Transmission Latency*: Over a conventional 9600 bps UHF link, the transmission of these 20,000 bits ($2,500 \text{ Bytes} \times 8$) requires 2.08 seconds.
- *Processing Latency*: Including an estimated 0.4 seconds for the onboard RISC-V core (@ 50MHz) to verify the Ed25519 signature, the total delay is approximately 2.48 seconds (rounded to 3s).

- *System Impact*: In the context of a standard 600-second (10-minute) LEO communication window, this represents an overhead of less than 0.5%, an optimal trade-off for ensuring immunity against unauthorized code injection.

D. Security Analysis

1) *Enhanced Threat Mitigation*: Our integrated approach addresses threats that bypass individual layers:

- **Compromised Ground Station**: Even with stolen signing keys, attackers cannot create valid firmware for specific hardware instances due to the PUF-based binding.
- **Software Vulnerability Exploit**: Code injection attacks (e.g., buffer overflows) fail to execute because the injected malicious binary does not match the target’s unique instruction set mapping.
- **Physical Attack on Satellite**: Scan chain obfuscation and the use of SRAM-PUFs prevent the extraction of secret keys even if the hardware is physically captured.

2) *Formal Security Properties*: The system provides:

- **Code Integrity**: $\forall$ firmware F , if F executes, then F was authorized for this specific hardware instance.
- **Forward Secrecy**: Compromise of current session keys does not affect the confidentiality of past communications.
- **Crypto Agility**: Cryptographic algorithms and ISA mappings can be updated in orbit without requiring hardware replacement.

VI. DISCUSSION

A. Security Improvements Over Current Approaches

Our integrated approach provides several key security improvements over existing solutions. First, it eliminates single points of failure (SPoF) by ensuring that compromising one layer (e.g., network encryption keys) does not compromise the entire system. Second, it provides defense-in-depth, with each layer providing complementary protections. Third, it enables cryptographic agility, allowing security parameters to be updated in response to new threats.

B. Applicability to CubeSat Constraints

The proposed solution is specifically designed to operate within CubeSat constraints. The total power overhead of approximately 7.2% is within acceptable limits for most CubeSat power budgets (2W–5W). The memory overhead of 12% is manageable given modern space-grade processors. The additional total delay of approximately 3 seconds per firmware update is negligible compared to typical 600-second communication windows. These trade-offs represent a reasonable balance between security and resource consumption.

C. Practical Implementation Considerations

Implementing the proposed architecture requires several practical considerations. First, ground station software must be modified to support hardware-specific firmware recompilation. Second, the CubeSat SoC must include the obfuscation logic,

which adds approximately 3.6% logic overhead (1,800 gates). Third, a key management infrastructure must be established to handle satellite-specific keys derived from the PUF (H_{id}).

D. Limitations and Open Challenges

Several limitations remain. First, the approach requires modifications to both ground and space segments. Second, the performance impact of instruction set diversification on compiler optimization needs further study. Third, the approach assumes a trusted initial provisioning phase. Future work should address these challenges through standardization and further hardware-level optimizations.

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